Solution:

For $X \sim U(a, b)$

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & otherwise \end{cases}$$

$P(20<X<40)=0.2$

$$\int_{20}^{40} \frac{1}{b-a} dx = 0.2$$

$$\frac{1}{b-a}(x)_{20}^{40} = 0.2$$

$$\frac{1}{b-a}(20) = 0.2$$

$$b - a = 100 \quad \ldots (1)$$

$$\text{mean} = 150$$

$$\frac{a+b}{2} = 150$$

$$a+b = 300 \quad \ldots (2)$$

(1)+(2) implies $2b = 400$ and $b = 200$.

Substituting b in (2) we have $a + 200 = 300$ and that $a = 100$.

a=100, b=200

$X \sim U(100,200)$

Example 10.20

If X is a Uniform variable U(a,b) with first and third quartiles 100 and 200, find the p.d.f of X.

Solution :

$$Q_1 = 100$$

$$\frac{3a+b}{4} = 100$$

$$3a+b = 400 \quad \ldots (1)$$

$$Q_3 = 200$$

$$\frac{a+3b}{4} = 200$$

$$a+3b = 800 \quad \ldots (2)$$